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Sylvester's criterion

In mathematics, **Sylvester's criterion** is a necessary and sufficient criterion to determine whether a Hermitian matrix is positive-definite.

Sylvester's criterion states that a $n \times n$ Hermitian matrix M is positive-definite if and only if all the following matrices have a positive determinant:

- the upper left 1-by-1 corner of M,
- the upper left 2-by-2 corner of M,
- the upper left 3-by-3 corner of M,
- $\vdots$
- M itself.

In other words, all of the *leading* principal minors must be positive. By using appropriate permutations of rows and columns of M, it can also be shown that the positivity of *any* nested sequence of n principal minors of M is equivalent to M being positive-definite.[1]

An analogous theorem holds for characterizing positive-semidefinite Hermitian matrices, except that it is no longer sufficient to consider only the *leading* principal minors as illustrated by the Hermitian matrix

$$\begin{pmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{pmatrix} \quad \textbf{with eigenvectors} \quad \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad \textbf{and} \quad \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

A Hermitian matrix M is positive-semidefinite if and only if *all* principal minors of M are nonnegative.[2][3]

Proof for the case of positive definite matrices

Suppose M_n is $n \times n$ Hermitian matrix $M_n^\dagger = M_n$. Let $M_k, k = 1, \dots n$ be the leading principal minor matrices, i.e. the $k \times k$ upper left corner matrices. It will be shown that if M_n is positive definite, then the principal minors are positive; that is, $\det M_k > 0$ for all k.

M_k is positive definite. Indeed, choosing

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_k \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} \vec{x} \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

we can notice that $0 < x^\dagger M_n x = \vec{x}^\dagger M_k \vec{x}$. Equivalently, the eigenvalues of M_k are positive, and this implies that $\det M_k > 0$ since the determinant is the product of the eigenvalues.

To prove the reverse implication, we use induction. The general form of an $(n+1) \times (n+1)$ Hermitian matrix is

$$M_{n+1} = \begin{pmatrix} M_n & \vec{v} \\ \vec{v}^\dagger & d \end{pmatrix} \qquad (*),$$

where M_n is an $n \times n$ Hermitian matrix, $\vec{v}$ is a vector and d is a real constant.

Suppose the criterion holds for M_n. Assuming that all the principal minors of M_{n+1} are positive implies that $\det M_{n+1} > 0$, $\det M_n > 0$, and that M_n is positive definite by the inductive hypothesis. Denote

$$x = \begin{pmatrix} \vec{x} \\ x_{n+1} \end{pmatrix}$$

then

$$x^\dagger M_{n+1} x = \vec{x}^\dagger M_n \vec{x} + x_{n+1} \vec{x}^\dagger \vec{v} + \bar{x}_{n+1} \vec{v}^\dagger \vec{x} + d|x_{n+1}|^2$$

By completing the squares, this last expression is equal to

$$\begin{aligned} &(\vec{x}^\dagger + \vec{v}^\dagger M_n^{-1} \bar{x}_{n+1}) M_n (\vec{x} + x_{n+1} M_n^{-1} \vec{v}) - |x_{n+1}|^2 \vec{v}^\dagger M_n^{-1} \vec{v} + d|x_{n+1}|^2 \\ &= (\vec{x} + \vec{c})^\dagger M_n (\vec{x} + \vec{c}) + |x_{n+1}|^2 (d - \vec{v}^\dagger M_n^{-1} \vec{v}) \end{aligned}$$

where $\vec{c} = x_{n+1} M_n^{-1} \vec{v}$ (note that M_n^{-1} exists because the eigenvalues of M_n are all positive.) The first term is positive by the inductive hypothesis. We now examine the sign of the second term. By using the block matrix determinant formula

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det A \det(D - CA^{-1}B)$$

on $(*)$ we obtain

$\det M_{n+1} = \det M_n (d - \vec{v}^\dagger M_n^{-1} \vec{v}) > 0$, which implies $d - \vec{v}^\dagger M_n^{-1} \vec{v} > 0$.

Consequently, $x^\dagger M_{n+1} x > 0$.

Proof for the case of positive semidefinite matrices

Let M_n be an n x n Hermitian matrix. Suppose M_n is semidefinite. Essentially the same proof as for the case that M_n is strictly positive definite shows that all principal minors (not necessarily the leading principal minors) are non-negative.

For the reverse implication, it suffices to show that if M_n has all non-negative principal minors, then for all $t>0$, all leading principal minors of the Hermitian matrix $M_n + tI_n$ are strictly positive, where I_n is the nxn identity matrix. Indeed, from the positive definite case, we would know that the matrices $M_n + tI_n$ are strictly positive definite. Since the limit of positive definite matrices is always positive semidefinite, we can take $t \to 0$ to conclude.

To show this, let M_k be the kth leading principal submatrix of M_n. We know that $q_k(t) = \det(M_k + tI_k)$ is a polynomial in t, related to the characteristic polynomial p_{M_k} via $q_k(t) = (-1)^k p_{M_k}(-t)$. We use the identity in Characteristic polynomial#Properties to write

$q_k(t) = \sum_{j=0}^{k} t^{k-j} \operatorname{tr}\left(\bigwedge^j M_k\right)$, where $\operatorname{tr}\left(\bigwedge^j M_k\right)$ is the trace of the jth exterior power of M_k.

From Minor_(linear_algebra)#Multilinear_algebra_approach, we know that the entries in the matrix expansion of $\bigwedge^j M_k$ (for $j > 0$) are just the minors of M_k. In particular, the diagonal entries are the principal minors of M_k, which of course are also principal minors of M_n, and are thus non-negative. Since the trace of a matrix is the sum of the diagonal entries, it follows that $\operatorname{tr}\left(\bigwedge^j M_k\right) \geq 0$. Thus the coefficient of t^{k-j} in $q_k(t)$ is non-negative for all $j > 0$. For $j = 0$, it is clear that the coefficient is 1. In particular, $q_k(t) > 0$ for all $t > 0$, which is what was required to show.

Notes

1. Horn, Roger A.; Johnson, Charles R. (1985), *Matrix Analysis*, Cambridge University Press, ISBN 978-0-521-38632-6. See Theorem 7.2.5.
2. Carl D. Meyer, Matrix Analysis and Applied Linear Algebra. See section **7.6 Positive Definite Matrices**, page 566
3. Prussing, John E. (1986), "The Principal Minor Test for Semidefinite Matrices" (https://web.archive.org/web/20170107084552/http://prussing.ae.illinois.edu/semidef.pdf) (PDF), *Journal of Guidance, Control, and Dynamics*, **9** (1): 121–122, Bibcode:1986JGCD....9..121P (https://ui.adsabs.harvard.edu/abs/1986JGCD....9..121P), doi:10.2514/3.20077 (https://doi.org/10.2514%2F3.20077), archived from the original (http://prussing.ae.illinois.edu/semidef.pdf) (PDF) on 2017-01-07, retrieved 2017-09-28

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- Horn, Roger A.; Johnson, Charles R. (1985), *Matrix Analysis*, Cambridge University Press, ISBN 978-0-521-38632-6. Theorem 7.2.5.
- Carl D. Meyer (June 2000), *Matrix Analysis and Applied Linear Algebra*, SIAM, ISBN 0-89871-454-0.

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